MTH314

Examination - July Semester 2025

Real Analysis II

Tuesday, 18 November 2025 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **SIX (6)** questions and comprises **SEVEN (7)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

MTH314 Copyright © 2025 Singapore University of Social Sciences (SUSS)

(Full marks: 100)

Question 1a

Define a function

$$f(x) = \frac{x}{\sqrt{x}+1}.$$

Show using only the limit definition of the derivative, and without L'Hospital's Rule, that for any $a > 0$,

$$f'(a) = \frac{\sqrt{a}+2}{2(\sqrt{a}+1)^2}.$$

(8 marks)

Question 1b

Let $f : \mathbb{R} \to \mathbb{R}$ be a twice-differentiable function such that $f(1) = 3$, $f(3) = 8$, $f(4) = 10$, $f(6) = 15$. Show that there exists some $c \in \mathbb{R}$ such that $f''(c) = 0$.

(7 marks)

Question 1c

Give an example of a twice-differentiable function $g : \mathbb{R} \to \mathbb{R}$ such that for every positive integer n, there exists exactly one real number x_n such that $g''(x_n) = n$, and provide an explicit formula for x_n in terms of n.

(3 marks)

Question 2

Let r be a positive real number. Suppose that $f : [3, 6] \to \mathbb{R}$ is a function that is three times differentiable on the closed interval $[3, 6]$ and which satisfies $f(4) = 16\ln(5)$,

$$f'(4) = \frac{16}{5} + 8\ln(5), f''(4) = \frac{64}{25} + 2\ln(5),$$

and $|f'''(x)| \leq r$ for $3 \leq x \leq 6$. Here, $\ln(w)$ denotes the natural logarithm of w, that is, $\log_e(w)$.

Question 2a

Find the Taylor polynomial $P_2(x)$ of $f(x)$ about the point $x = 4$, that is, the Taylor expansion about 4 up to and including the term $(x-4)^2$.

(4 marks)

Question 2b

Find the maximum error in estimating $f(x)$ in the domain $[3, 6]$ using the Taylor polynomial $P_2(x)$ about $x = 4$, expressing your answer in terms of r.

(3 marks)

Question 2c

Use an appropriate Taylor polynomial to estimate the value of

$$\int_4^5 f(x)\, dx,$$

giving your answer to 4 decimal place accuracy.

(4 marks)

Question 2d

Define a function $g : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$g(x) = x^2 \ln(x+1).$$

Question 2d(i)

Prove that

$$g(4) = 16\ln(5), g'(4) = \frac{16}{5} + 8\ln(5), g''(4) = \frac{64}{25} + 2\ln(5).$$

(5 marks)

Question 2d(ii)

Given that

$$\int g(x)\, dx = \frac{1}{18}(6(x^3+1)\ln(x+1) - 2x^3 + 3x^2 - 6x) + C,$$

determine the percentage error that would be incurred,

$$\frac{\text{difference between actual and estimated value}}{\text{actual value}} \times 100\%,$$

in estimating

$$\int_4^5 g(x)\, dx$$

using the Taylor polynomial $P_2(x)$ of $g(x)$ about the point $x = 4$ compared to integrating the actual function.

(4 marks)

Question 3a

Let $a, b \in \mathbb{R}$. Define a function $g : \mathbb{R} \to \mathbb{R}$,

$$g(x) = \begin{cases} 2\cos(x)\ , & x \leq 0 \\ x^4 + ax + b(2^x)\ , & x > 0 \end{cases}$$

Determine the values of a, b such that g is differentiable on $\mathbb{R}$.

(8 marks)

Question 3b

Let n, m be positive integers, where $n \geq m$. Define a function $f : [0, \infty) \to \mathbb{R}$ by

$$f(x) = x^2 + n\pi x + \sin(mx).$$

Show that f is strictly increasing for $x \geq 0$, and compute

$$(f^{-1})'(3\pi^2(n+3)),$$

expressing your final answer in terms of n, m.

(8 marks)

Question 4

Define a function $f : [-3, 4] \to \mathbb{R}$,

$$f(x) = 4x^3 - 3x^2 - 18x + 1.$$

Define a tagged partition $\mathcal{P}$ of $[-3, 4]$ comprising of the following intervals and associated tags:

$$([-3, -2], -2.5), ([-2, -1.3], -1.7), ([-1.3, -0.2], -1),$$
$$([-0.2, 1], 0.3), ([1, 1.8], 1.5), ([1.8, 3], 2), ([3, 4], 3).$$

Question 4a

Show that f has a local maximum point at $x = -1$ and a local minimum point at $x = 1.5$.

(6 marks)

Question 4b

Calculate the value of the Riemann Sum $S(f, \mathcal{P})$.

(4 marks)

Question 4c

Let P denote the untagged partition with the same intervals of $\mathcal{P}$. Calculate the value of the Lower Riemann Sum $L(f, P)$ and the Upper Riemann Sum $U(f, P)$.

(8 marks)

Question 5

Given $a, b \in \mathbb{R}$ with $a < b$, let $f : [a, b] \to \mathbb{R}$ be a non-constant function that is continuous on $[a, b]$. It is known that for such a function, given any w in the range of f that is not the maximum of f in $[a, b]$, there exists a finite collection of non-empty, pairwise disjoint closed subintervals $(I_r \ : \ 1 \leq r \leq k)$ of $[a, b]$ such that $f(x) \geq w$ for any $w \in I_r$, $1 \leq r \leq k$, and $f(x) < w$ for any $x \in [a, b]$ that is not within any of these intervals. Let $M = \sup\{|f(x)| \ : \ a \leq x \leq b\}$.

Question 5a

Show that

$$\lim_{n\to\infty} \left(\int_a^b |f(x)|^n \, dx \right)^{1/n}$$

exists and is equal to M.

(11 marks)

Question 5b

Let $a \in [1, \infty)$. Use the result in Question 5a to derive the inequality

$$\lim_{n\to\infty} \left(\int_0^1 x^a e^{-nx^2} \, dx \right)^{1/n} \geq (2e)^{-1/2}.$$

(7 marks)

Question 6

Show that the limit

$$\lim_{n\to\infty} \sum_{i=1}^{n} \frac{\ln(2n+i) - \ln(n)}{2n+i}$$

can be transformed into a definite integral and compute the limit by evaluating the definite integral.

(10 marks)

----- END OF PAPER -----